

West African Crime Corrected

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Introduction ## This project analyses major crime data from selected cities in West African countries collected in 1974. Using R, I created and explored vectors containing the countries, cities, and number of crimes. I then organized the data into a data frame, calculated the total number of crimes, classified cities according to their crime rates, and compared crime totals between countries. To approach this we make use of Base R function and dplyr package()



Figure 1: crime rate



Figure 2: crime rate

```
# Question 1a  
# Create a vector containing all countries shown in the table 1  
# Table 1: Count of the number of crimes in cities in West African
```

```
country <- c("Nigeria", "Nigeria", "Nigeria", "Cameroon", "Nigeria", "Nigeria", "Nigeria", "C  
"Nigeria", "Nigeria", "Nigeria", "Nigeria", "Nigeria", "Nigeria", "Cameroon", "Nigeria", "Ni  
country
```

```
[1] "Nigeria" "Nigeria" "Nigeria" "Cameroon" "Nigeria" "Nigeria"  
[7] "Nigeria" "Cameroon" "Nigeria" "Nigeria" "Nigeria" "Nigeria"  
[13] "Nigeria" "Nigeria" "Nigeria" "Nigeria" "Cameroon" "Nigeria"  
[19] "Nigeria" "Nigeria"
```

```
length(country)
```

```
[1] 20
```

```
# 1c  
length(unique(country))
```

```
[1] 2
```

```
# 1c  
unique(country)
```

```
[1] "Nigeria" "Cameroon"
```

```
# 1d  
city <- c("Maiduguri", "Unknown", "Kano", "Bamenda", "Port Harcourt", "Potiskum", "Damaturu",  
# 1d  
city
```

```
[1] "Maiduguri" "Unknown" "Kano" "Bamenda"  
[5] "Port Harcourt" "Potiskum" "Damaturu" "Unknown"  
[9] "Jos" "Lagos" "Gombe" "Kaduna"  
[13] "Logo district" "Konduga" "Warri" "Bama"  
[17] "Buea" "Buni Yadi" "Damboa" "Gamboru"
```

```
# 1d  
length(city)
```

```
[1] 20
```

```
length(unique(city))
```

```
[1] 19
```

```
unique(city)
```

```
[1] "Maiduguri" "Unknown" "Kano" "Bamenda"  
[5] "Port Harcourt" "Potiskum" "Damaturu" "Jos"  
[9] "Lagos" "Gombe" "Kaduna" "Logo district"  
[13] "Konduga" "Warri" "Bama" "Buea"  
[17] "Buni Yadi" "Damboa" "Gamboru"
```

```
# 1d
n_crime <- c( 488, 225, 130, 87, 76,
             65, 63, 55, 51, 46,
             42, 38, 37, 36, 35,
             32, 31, 30, 30, 29)
n_crime
```

```
[1] 488 225 130 87 76 65 63 55 51 46 42 38 37 36 35 32 31 30 30
[20] 29
```

```
length(n_crime)
```

```
[1] 20
```

```
length(unique(n_crime))
```

```
[1] 19
```

```
unique(n_crime)
```

```
[1] 488 225 130 87 76 65 63 55 51 46 42 38 37 36 35 32 31 30 29
```

```
# 1e
class(country)
```

```
[1] "character"
```

```
class(city)
```

```
[1] "character"
```

```
class(n_crime)
```

```
[1] "numeric"
```

```
# 1e  
str(country)
```

```
chr [1:20] "Nigeria" "Nigeria" "Nigeria" "Cameroon" "Nigeria" "Nigeria" ...
```

```
str(city)
```

```
chr [1:20] "Maiduguri" "Unknown" "Kano" "Bamenda" "Port Harcourt" ...
```

```
str(n_crime)
```

```
num [1:20] 488 225 130 87 76 65 63 55 51 46 ...
```

```
# 1e  
is.character(country)
```

```
[1] TRUE
```

```
is.character(city)
```

```
[1] TRUE
```

```
is.numeric(n_crime)
```

```
[1] TRUE
```

```
# 1e  
is.vector(country)
```

```
[1] TRUE
```

```
is.vector(city)
```

```
[1] TRUE
```

```
is.vector(n_crime)
```

```
[1] TRUE
```

```
# 1f  
sum(n_crime)
```

```
[1] 1626
```

```
# 1g  
# Ans: Data frame  
  
# 1h  
crime_data <- data.frame(country, city, n_crime)  
crime_data
```

	country	city	n_crime
1	Nigeria	Maiduguri	488
2	Nigeria	Unknown	225
3	Nigeria	Kano	130
4	Cameroon	Bamenda	87
5	Nigeria	Port Harcourt	76
6	Nigeria	Potiskum	65
7	Nigeria	Damaturu	63
8	Cameroon	Unknown	55
9	Nigeria	Jos	51
10	Nigeria	Lagos	46
11	Nigeria	Gombe	42
12	Nigeria	Kaduna	38
13	Nigeria	Logo district	37
14	Nigeria	Konduga	36
15	Nigeria	Warri	35
16	Nigeria	Bama	32
17	Cameroon	Buea	31
18	Nigeria	Buni Yadi	30
19	Nigeria	Damboa	30
20	Nigeria	Gamboru	29

```
# 1i
crime_rate <- character(length(n_crime))
crime_rate
```

```
[1] "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""
```

```
crime_data <- data.frame(country, city, n_crime, crime_rate)
crime_data
```

	country	city	n_crime	crime_rate
1	Nigeria	Maiduguri	488	
2	Nigeria	Unknown	225	
3	Nigeria	Kano	130	
4	Cameroon	Bamenda	87	
5	Nigeria	Port Harcourt	76	
6	Nigeria	Potiskum	65	
7	Nigeria	Damaturu	63	
8	Cameroon	Unknown	55	
9	Nigeria	Jos	51	
10	Nigeria	Lagos	46	
11	Nigeria	Gombe	42	
12	Nigeria	Kaduna	38	
13	Nigeria	Logo district	37	
14	Nigeria	Konduga	36	
15	Nigeria	Warri	35	
16	Nigeria	Bama	32	
17	Cameroon	Buea	31	
18	Nigeria	Buni Yadi	30	
19	Nigeria	Damboa	30	
20	Nigeria	Gamboru	29	

```
crime_data$crime_rate <- ifelse(
  test = crime_data$n_crime >= 400,
  yes = "High",
  no = ifelse(
    test = crime_data$n_crime >= 100 & crime_data$n_crime <= 399,
    yes = "Medium",
    no = "Low"
  )
)
```

```
crime_data
```

	country	city	n_crime	crime_rate
1	Nigeria	Maiduguri	488	High
2	Nigeria	Unknown	225	Medium
3	Nigeria	Kano	130	Medium
4	Cameroon	Bamenda	87	Low
5	Nigeria	Port Harcourt	76	Low
6	Nigeria	Potiskum	65	Low
7	Nigeria	Damaturu	63	Low
8	Cameroon	Unknown	55	Low
9	Nigeria	Jos	51	Low
10	Nigeria	Lagos	46	Low
11	Nigeria	Gombe	42	Low
12	Nigeria	Kaduna	38	Low
13	Nigeria	Logo district	37	Low
14	Nigeria	Konduga	36	Low
15	Nigeria	Warri	35	Low
16	Nigeria	Bama	32	Low
17	Cameroon	Buea	31	Low
18	Nigeria	Buni Yadi	30	Low
19	Nigeria	Damboa	30	Low
20	Nigeria	Gamboru	29	Low

```
subset(crime_data, country == "Nigeria")
```

	country	city	n_crime	crime_rate
1	Nigeria	Maiduguri	488	High
2	Nigeria	Unknown	225	Medium
3	Nigeria	Kano	130	Medium
5	Nigeria	Port Harcourt	76	Low
6	Nigeria	Potiskum	65	Low
7	Nigeria	Damaturu	63	Low
9	Nigeria	Jos	51	Low
10	Nigeria	Lagos	46	Low
11	Nigeria	Gombe	42	Low
12	Nigeria	Kaduna	38	Low
13	Nigeria	Logo district	37	Low
14	Nigeria	Konduga	36	Low
15	Nigeria	Warri	35	Low

16	Nigeria	Bama	32	Low
18	Nigeria	Buni Yadi	30	Low
19	Nigeria	Damboa	30	Low
20	Nigeria	Gamboru	29	Low

```
aggregate(crime_data$n_crime, by = list(country = country), FUN = sum)
```

	country	x
1	Cameroon	173
2	Nigeria	1453

```
#
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
## Using dplyr
```

```
crime_data |>
  summarize(
    .by = country,
    total_crime = sum(n_crime),
    average_no_of_crime = mean(n_crime)
  )
```

	country	total_crime	average_no_of_crime
1	Nigeria	1453	85.47059
2	Cameroon	173	57.66667

```
crime_data |>
  summarize(
    .by = country,
    total_crime = sum(n_crime),
    average_no_of_crime = mean(n_crime)
  ) |>
  top_n( n = 1, wt =total_crime)
```

	country	total_crime	average_no_of_crime
1	Nigeria	1453	85.47059